

Mensuration — Area & Volume

COMPLETE LEARNING GUIDE

What You'll Learn

Mensuration questions test direct application of area, surface area, and volume formulas for standard 2D and 3D shapes, but the real scoring differentiator is recognizing which formula variant a word problem actually needs — a closed cylinder's surface area differs from an open one, a cone's slant height differs from its vertical height, and 'melt and recast' problems conserve volume while 'painting' problems care about surface area. This sub-module works through 2D shape areas and perimeters, circle and sector formulas, the volume and surface area of cuboids, cylinders, cones, and spheres, the scaling rule connecting linear dimensions to area and volume, composite solids built from multiple shapes, and classic practical word problems like paths around fields and melting-and-recasting — each with a complete worked numerical example using clean, verifiable numbers.

1. 2D Areas and Perimeters: Rectangle, Square, Triangle, and Trapezium

The basic 2D formulas — rectangle area = length×width, square area = side², triangle area = $\frac{1}{2} \times \text{base} \times \text{height}$, and trapezium area = $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$ — underlie nearly every 2D mensuration question, often combined with each other in composite figures.

- A rectangle 12 m long and 5 m wide has area = $12 \times 5 = 60 \text{ m}^2$ and perimeter = $2(12+5) = 34 \text{ m}$.
- A triangle with base 10 cm and height 6 cm has area = $\frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$.
- A trapezium with parallel sides 10 cm and 14 cm and height 5 cm has area = $\frac{1}{2} \times (10+14) \times 5 = \frac{1}{2} \times 24 \times 5 = 60 \text{ cm}^2$.

2. Circles: Circumference, Area, Sector, and Arc Length

Circumference = $2\pi r$, area = πr^2 For a sector spanning angle θ , Fraction of circle, sector area = $\frac{\theta}{360} \times \pi r^2$ and arc length = $\frac{\theta}{360} \times 2\pi r$ — both are simply the fraction of the full circle that the sector's angle represents.

- A circle has radius 7 cm (using $\pi = \frac{22}{7}$) Circumference = $2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$. Area = $\frac{22}{7} \times 7^2 = 154 \text{ cm}^2$.

- A 90° sector of the same circle: sector area = $(90/360) \times 154 = 38.5 \text{ cm}^2$. Arc length = $(90/360) \times 44 = 11 \text{ cm}$.

3. Surface Area and Volume of Cuboids and Cubes

Cuboid volume = length $\times$ width $\times$ height; total surface area = $2(lw + wh + hl)$, summing the areas of all 6 faces. A cube is the special case where all three dimensions are equal: volume = side³, surface area = $6 \times \text{side}^2$.

- A cuboid measuring 8 cm $\times$ 5 cm $\times$ 4 cm has volume = $8 \times 5 \times 4 = 160 \text{ cm}^3$ and surface area = $2(8 \times 5 + 5 \times 4 + 4 \times 8) = 2(40 + 20 + 32) = 2 \times 92 = 184 \text{ cm}^2$.
- A cube of side 5 cm has volume = $5^3 = 125 \text{ cm}^3$ and surface area = $6 \times 5^2 = 6 \times 25 = 150 \text{ cm}^2$.

4. Surface Area and Volume of Cylinders

Cylinder volume = $\pi r^2 h$; Curved (lateral) surface area = $2\pi r h$; Total surface area (including both circular ends) = $2\pi r^2 + 2\pi r h$ — use the curved surface area alone for open-ended objects like pipes, and the total surface area for a fully closed solid cylinder.

- A cylinder has radius 7 cm and height 10 cm ($\pi \approx 22/7$) Volume = $22/7 \times 49 \times 10 = 1540 \text{ cm}^3$. Curved surface area = $2 \times 22/7 \times 7 \times 10 = 440 \text{ cm}^2$. Total surface area (closed) = $440 + 2 \times 154 = 440 + 308 = 748 \text{ cm}^2$.

5. Surface Area and Volume of Cones and Spheres

Cone volume = $(1/3)\pi r^2 h$; Curved surface area = $\pi r l$, where l is the SLANT height ($l = \sqrt{r^2 + h^2}$), not the vertical height — these are different unless the question explicitly says so. Sphere volume = $(4/3)\pi r^3$; Surface area = $4\pi r^2$

- A cone has radius 7 cm and vertical height 24 cm. Slant height $l = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25 \text{ cm}$. Volume = $1/3 \times 22/7 \times 49 \times 24 = 1232 \text{ cm}^3$. Curved surface area = $22/7 \times 7 \times 25 = 550 \text{ cm}^2$.
- A sphere has radius 7 cm. Volume = $4/3 \times 22/7 \times 343 = 1437.33 \text{ cm}^3$. Surface area = $4 \times 22/7 \times 49 = 616 \text{ cm}^2$.

6. Effect of Scaling Dimensions on Area and Volume

When every linear dimension of a solid is scaled by a factor k , surface area scales by k^2 and volume scales by k^3 — NOT by k itself. This non-linear scaling is a frequent source of error when a question describes an enlarged or shrunk version of a shape.

- A cube of side 4 cm (volume 64 cm^3 , surface area 96 cm^2) is enlarged to side 8 cm — a scale factor $k=2$. New volume = $64 \times 2^3 = 64 \times 8 = 512 \text{ cm}^3$ (matches $8^3 = 512$). New surface area = $96 \times 2^2 = 96 \times 4 = 384 \text{ cm}^2$ (matches $6 \times 8^2 = 384$).

7. Composite Solids

For a solid built from two or more basic shapes joined together, compute each component's volume (or surface area) separately using its own formula, then add them — being careful not to double-count any surface where the shapes join.

- A solid consists of a cylinder (radius 7 cm, height 10 cm) topped with a hemisphere of the same radius. Total volume = cylinder volume + hemisphere volume = $1540 + (2/3 \times 22/7 \times 343) = 1540 + 718.67 = 2258.67 \text{ cm}^3$.

8. Practical Word Problems: Paths and Melting-Recasting

For a uniform-width path around a rectangular field, compute the OUTER area (field plus path on all sides) minus the field's own area. For melting-and-recasting problems, the VOLUME of metal is conserved (not its shape or surface area) — set the two shapes' volumes equal and solve for the unknown dimension.

- A rectangular field $30 \text{ m} \times 20 \text{ m}$ has a 2 m wide path built around its outside. Outer dimensions = $34 \text{ m} \times 24 \text{ m}$, outer area = 816 m^2 . Field area = 600 m^2 . Path area = $816 - 600 = 216 \text{ m}^2$.

- A metallic sphere of radius 6 cm is melted and recast into a cylinder of radius 4 cm . Find the cylinder's height. Sphere volume = $4/3 \times \pi \times 6^3$. Cylinder volume = $\pi \times 4^2 \times h$. Setting them equal: $16h = 288 \Rightarrow h = 18 \text{ cm}$.

Common Pitfalls

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- Using the curved surface area formula when the question actually needs total surface area (or vice versa) for a closed cylinder or similar solid.
 - Substituting diameter where radius is required (or vice versa) in circle, cylinder, cone, or sphere formulas.
 - Assuming volume scales the same way as area when dimensions change — area scales with k^2 , volume scales with k^3 , never linearly with k .
 - In melt-and-recast problems, equating surface areas instead of volumes — the quantity of metal (volume) is what's conserved, not the shape's surface area.
 - Forgetting the 'outer minus inner' approach for a path around a field, or accidentally computing the field's area minus the path instead of the reverse.
 - Confusing a cone's slant height (used for curved surface area) with its vertical height (used for volume) — they differ unless explicitly stated equal.
 - Using π too early in a calculation and π too late, causing rounding mismatches in the final answer.

Key Takeaways

- Circle: circumference = $2\pi r$, area = πr^2 ; sector area = $\frac{\theta}{360} \pi r^2$; arc length = $\frac{\theta}{360} 2\pi r$.
- Cuboid: volume = $l \times w \times h$; surface area = $2(lw + wh + hl)$. Cube: volume = side^3 ; surface area = $6 \times \text{side}^2$.
- Cylinder: volume = $\pi r^2 h$; curved surface area = $2\pi r h$; total surface area = $2\pi r (r + h)$.
- Cone: volume = $\frac{1}{3} \pi r^2 h$; curved surface area = $\pi r l$ where $l = \sqrt{r^2 + h^2}$ is the SLANT height, not vertical height.
- Sphere: volume = $\frac{4}{3} \pi r^3$; surface area = $4\pi r^2$.
- Scaling all linear dimensions by k multiplies area by k^2 and volume by k^3 .
- Melt-and-recast problems conserve VOLUME, not surface area or shape — set the two volumes equal and solve.

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GeeksforGeeks — Mensuration Formulas

<https://www.geeksforgeeks.org/maths/mensuration/>

IndiaBix — Area Aptitude Questions

<https://www.indiabix.com/aptitude/area/>

IndiaBix — Volume and Surface Area Aptitude Questions

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